Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal | Lecture 41

Clinical Anatomy of the Urinary System – Flipped Class

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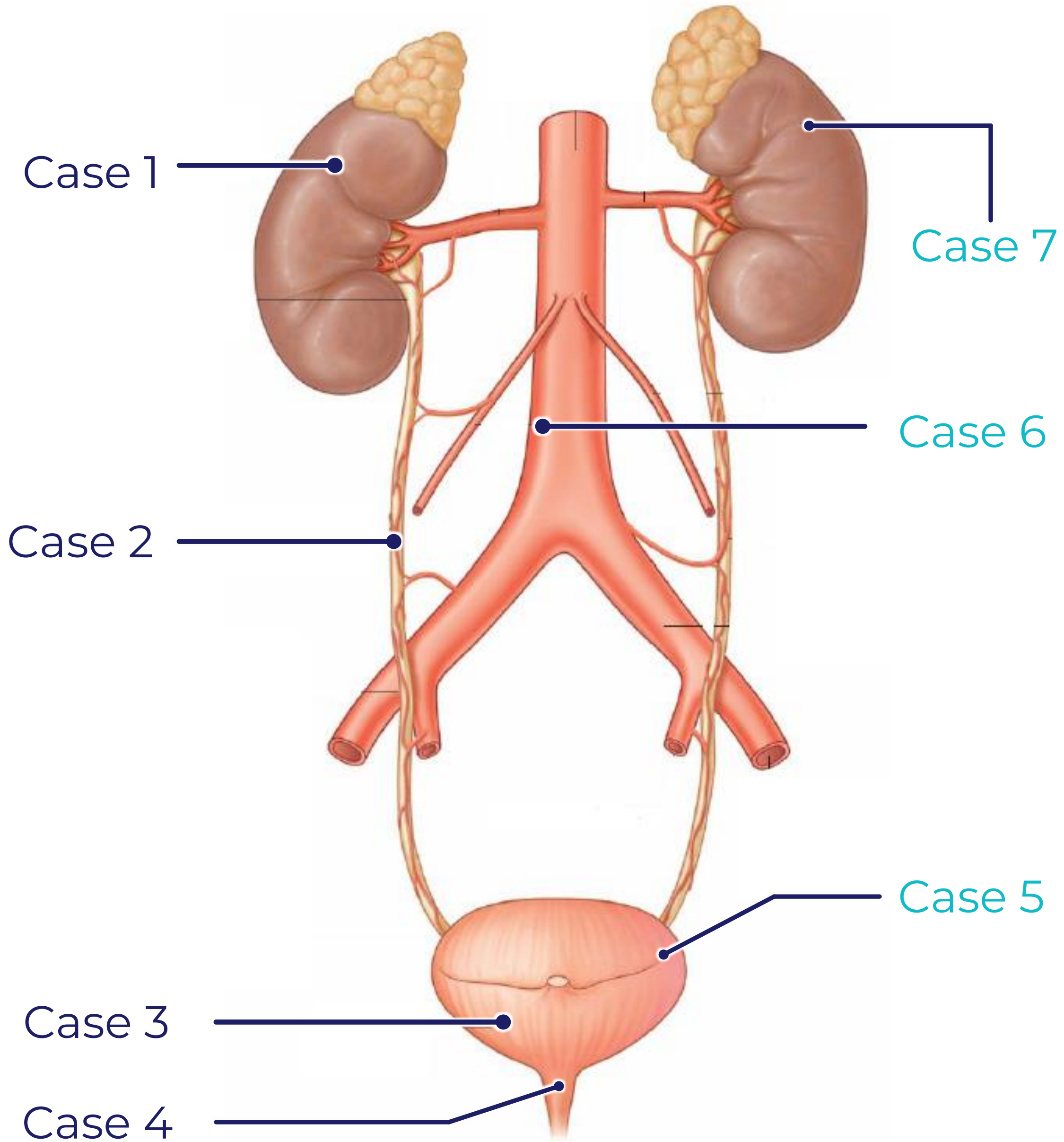
Objectives

SOM.MK.II.BPM1.3.CPR.2.ANAT.1474	Describe ''the water under the bridge'' relationship
SOM.MK.II.BPM1.3.CPR.2.ANAT.1476	List the sites of constriction of the ureters and explain their relationship to kidney stones
SOM.MK.II.BPM1.3.CPR.2.ANAT.1480	Describe the flow of urine from the renal papillae to the ureter and to the urinary bladder.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1481	Describe the retropubic space.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1482	Compare and contrast the male versus female urethra.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1483	Compare the innervation of the internal and external urethral sphincters.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1484	Describe the muscles and nerves involved in normal micturition and urinary continence.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1487	Identify, in radiological images, the major organs of the renal system including the kidneys, ureters and bladder
SOM.MK.II.BPM1.3.CPR.2.ANAT.1488	Define the term "cystocele" and explain why it is more common in multiparous women.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1489	Define the term "stress urinary incontinence" and explain which structures may be injured during vaginal delivery to lead to this condition.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1491	Explain the distribution of pain from kidney stones that are impacted (lodged) in the ureter.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1492	Describe the layers that would be cut when harvesting a donor kidney using a posterior approach.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1493	Describe the anatomical location, mechanism and clinical consequences of renal vein entrapment syndrome.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1494	Describe the development of cystic kidney disease.

Overview of Cases



A 30-year-old man is brought to the emergency department after being stabbed in the back during an altercation. The stab wound is located in the left posterior lumbar region, just below the 12th rib and lateral to the vertebral column. He is hemodynamically stable but reports flank pain and visible blood in his urine. Which of the following best describes the correct sequence of structures that the knife blade would have pierced in a posterior approach to reach the kidney?

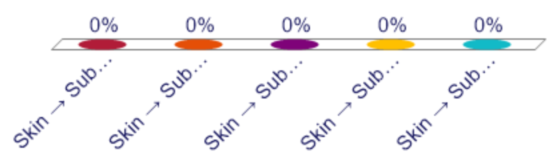
A. Skin → Subcutaneous tissue → Latissimus dorsi → Internal oblique → Transversus abdominis → Renal capsule

B. Skin → Subcutaneous tissue → Latissimus dorsi → Serratus anterior → Thoracolumbar fascia → Kidney

C. Skin → Subcutaneous tissue → Thoracolumbar fascia → Latissimus dorsi → Quadratus lumborum → Renal capsule

D. Skin → Subcutaneous tissue → Latissimus dorsi → Erector spinae → Thoracolumbar fascia → Quadratus lumborum → Kidney

E. Skin → Subcutaneous tissue → Latissimus dorsi → Thoracolumbar fascia → Internal oblique → Renal fascia → Pararenal fat



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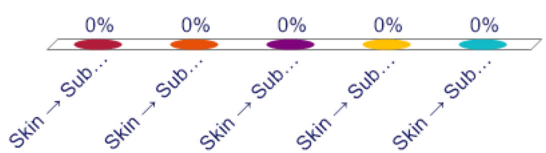

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Case 1

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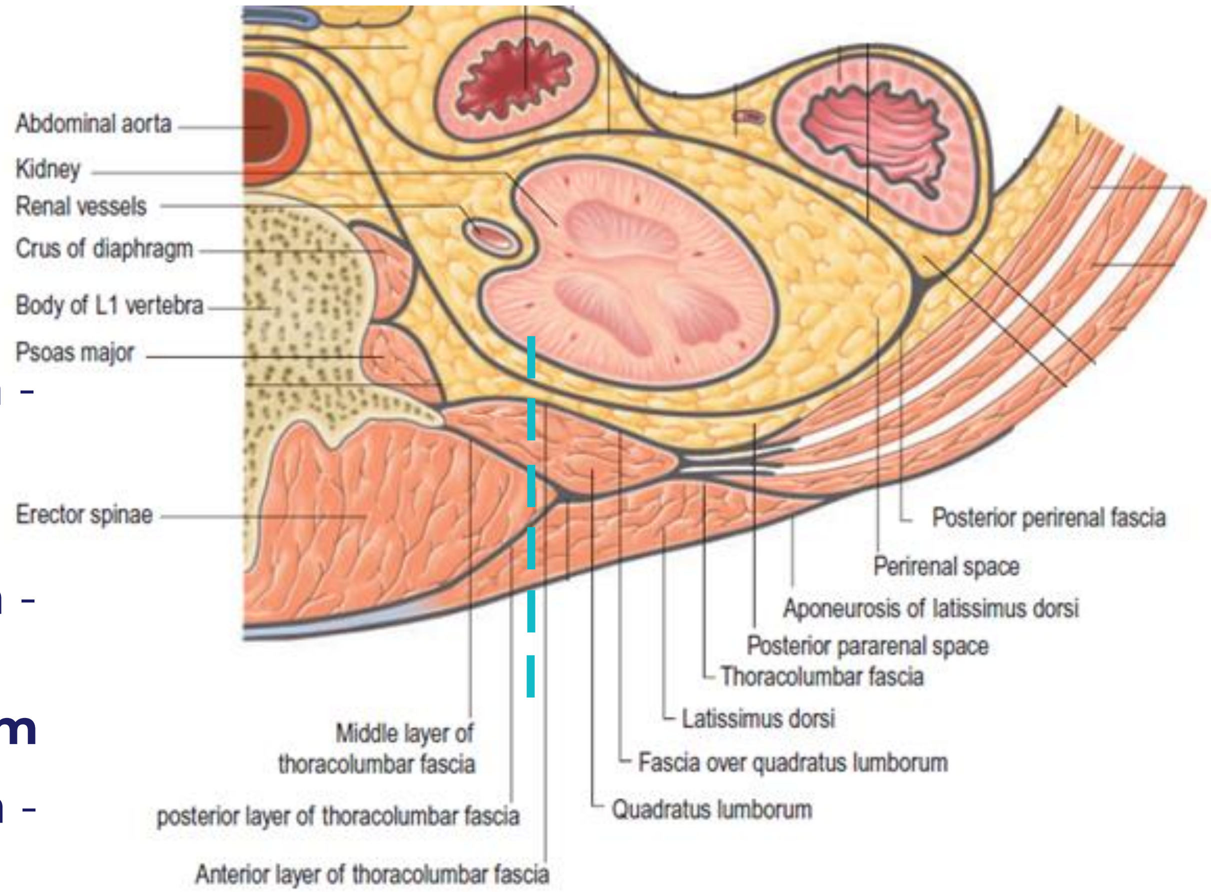
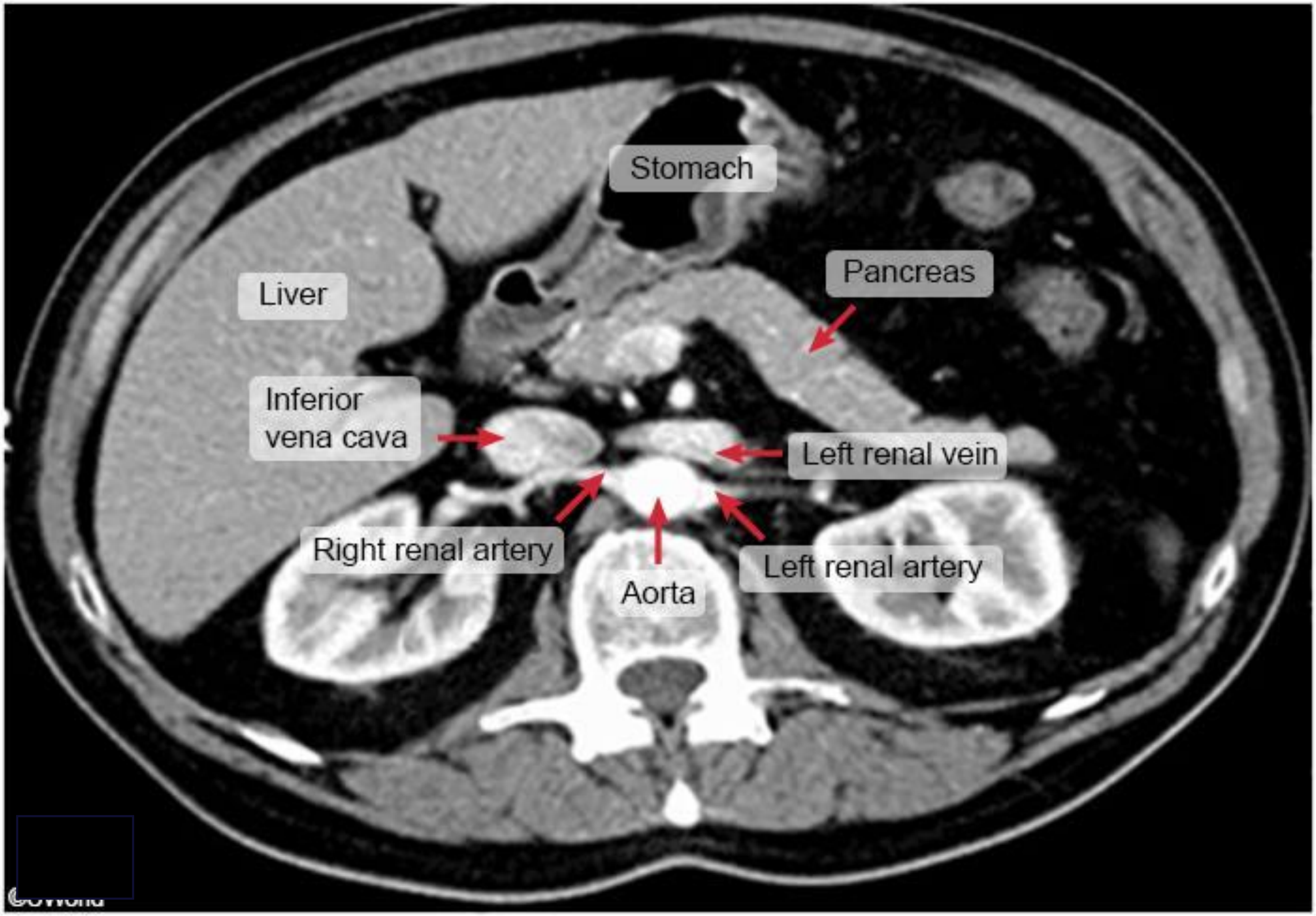
- What would be appropriate imaging modalities for monitoring this patient?
- What anatomical layers would a knife penetrate when entering the posterior lumbar region to reach the kidney?

Posterior Approach to the Kidneys

1. Skin
2. Subcutaneous tissue
3. **Latissimus dorsi**
4. Serratus posterior inferior *(not shown)*
5. Thoracolumbar fascia - posterior layer
6. **Erector spinae**
7. Thoracolumbar fascia - middle layer
8. **Quadratus lumborum**
9. Thoracolumbar fascia - anterior layer
10. **Pararenal fat**
11. **Renal Fascia**
12. **Perirenal fat**

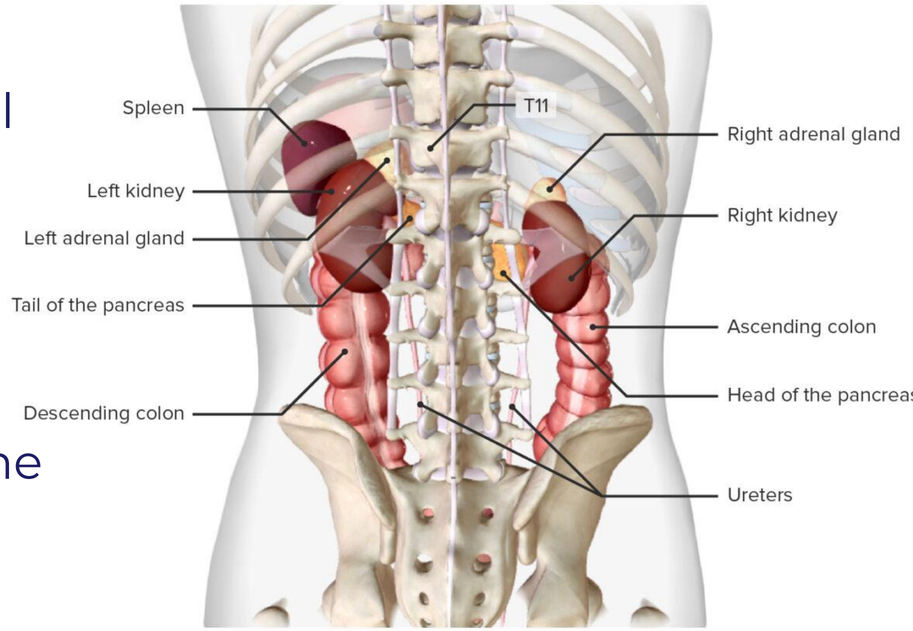


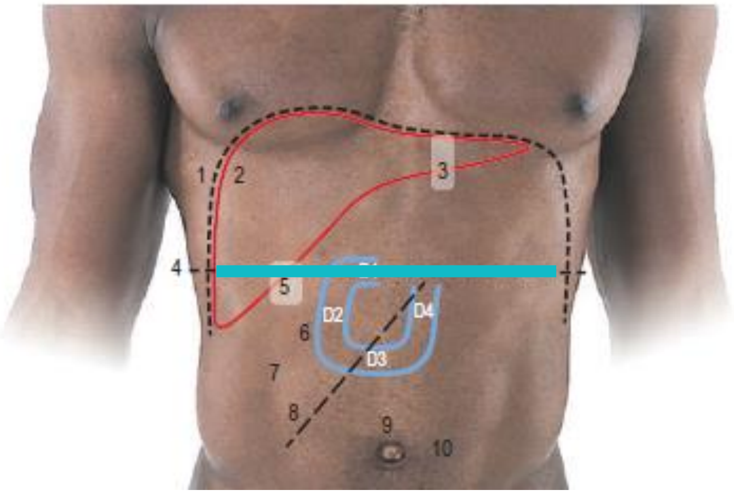
Imaging - CT

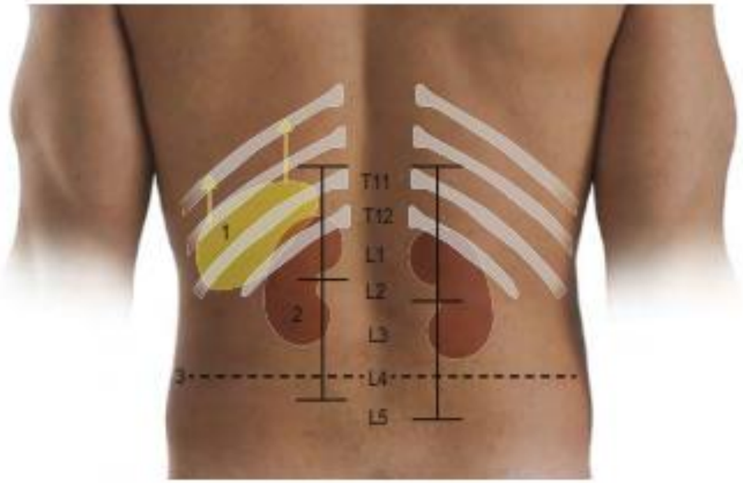

Stomach
Pancreas
Liver
Inferior vena cava
Left renal vein
Right renal artery
Left renal artery
Aorta

Kidneys

- Transpyloric plane
    - Horizontal line at the L1 level
    - Midway between jugular notch and top of pubic symphysis
    - Right kidney lies just below the plane
    - Left kidney lies just above the plane
- Hilum is approximately 5 cm from midline



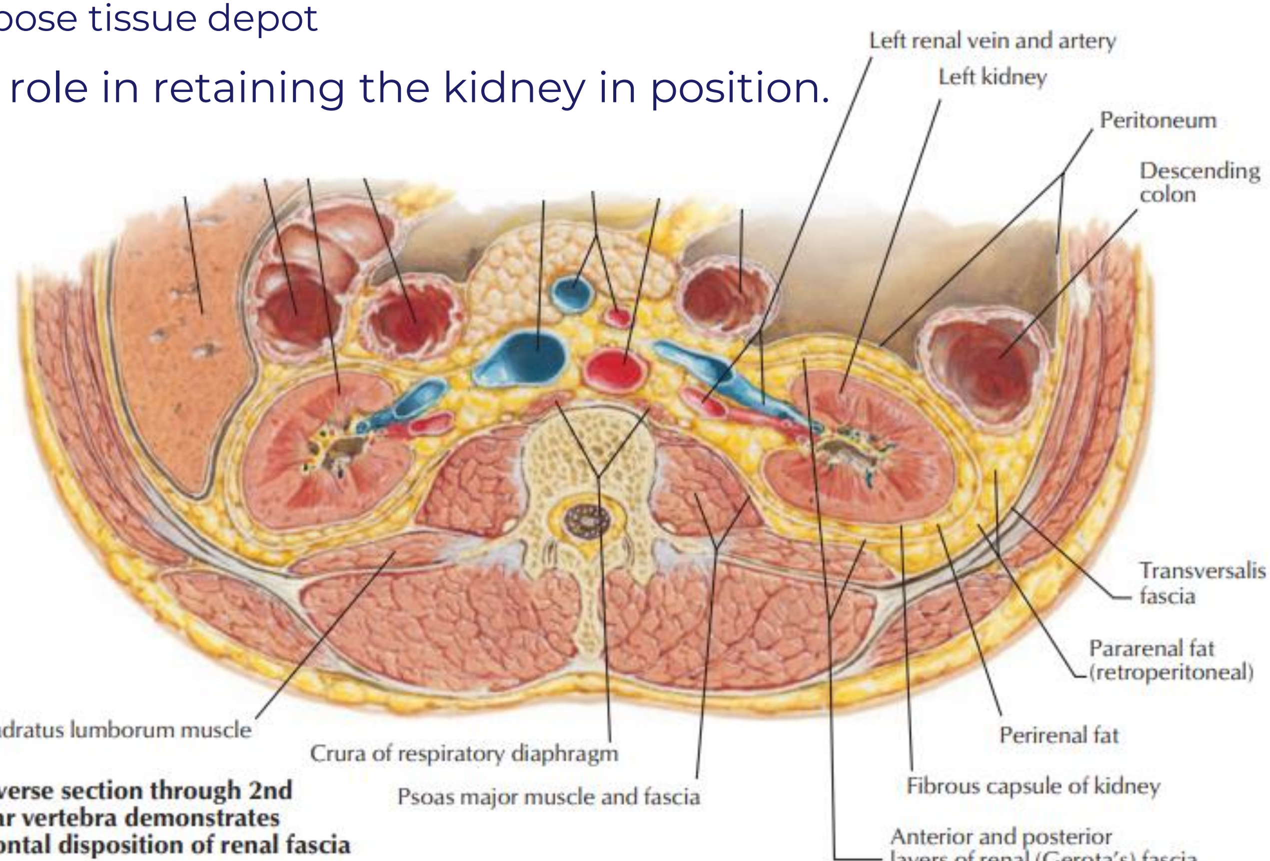




Renal Fat and Fascia

- Perinephric (perirenal) fat): outside the renal capsule
  - clinical predictor for many comorbidities
- Renal fascia (of Gerota): between parietal peritoneum & posterior abdominal wall
  - restrains the extension of perinephric abscesses
- Paranephric (pararenal) fat: extraperitoneal
  - white adipose tissue depot

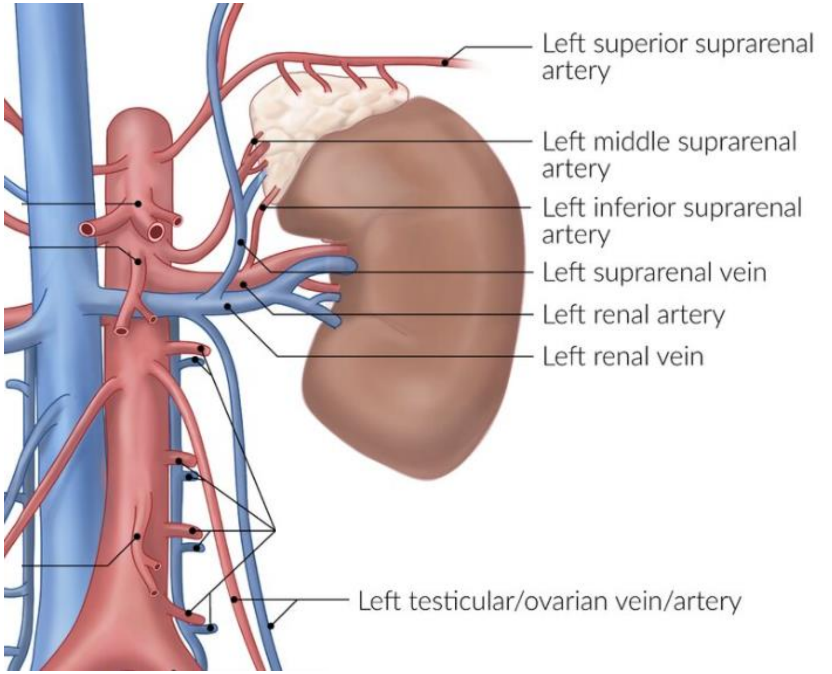
All layers play a role in retaining the kidney in position.



Transverse section through 2nd lumbar vertebra demonstrates horizontal disposition of renal fascia

Renal transplant
Donor Kidney (Usually Left)

- Divide adrenal vein
    - Dissect adrenal gland from upper pole
- Divide gonadal vein
    - Can be used to facilitate dissection of ureter
- Dissect renal artery at level of the aorta



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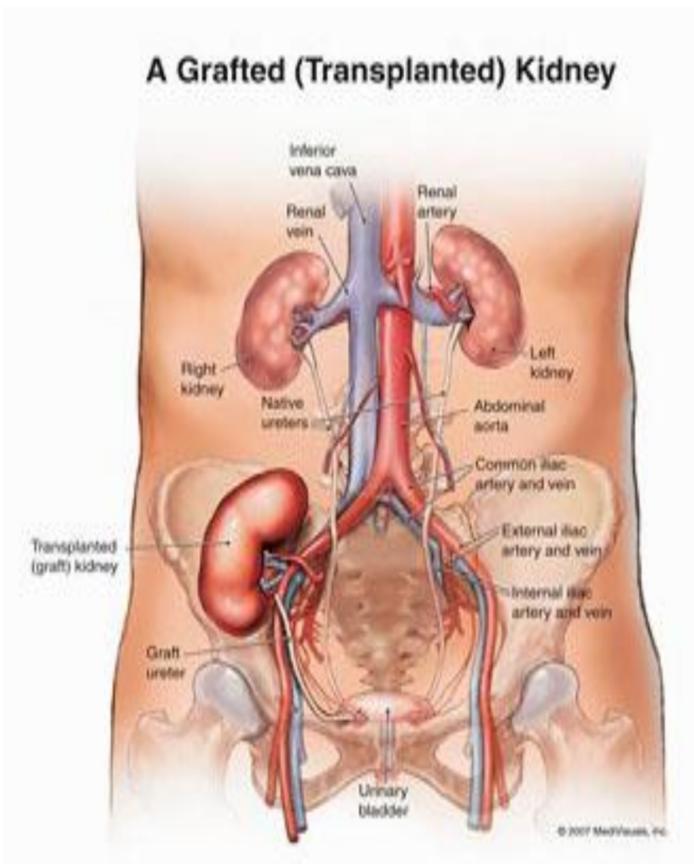
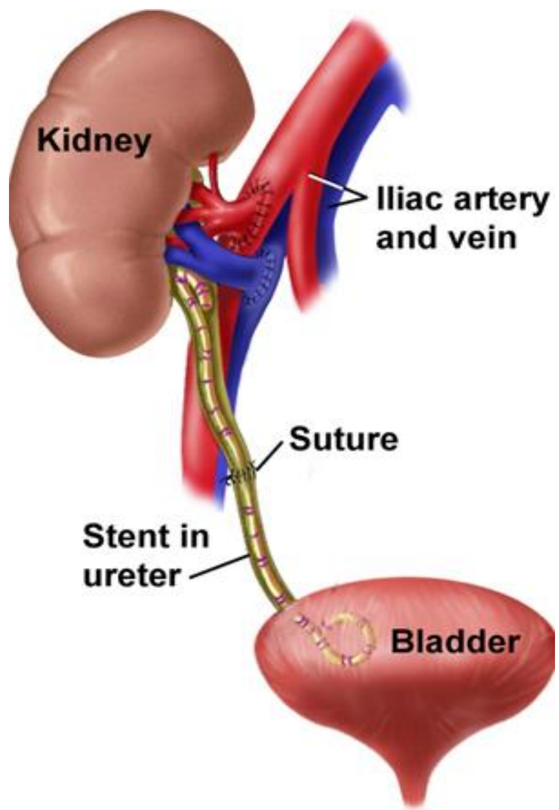
Renal transplant
Graft Recipient

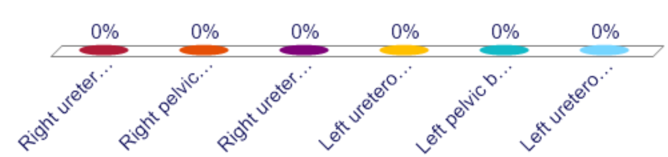
- Transplant is attached in the iliac fossa
- Renal artery/vein are joined to External iliac artery/vein
- Ureter is sutured into the urinary bladder





A 22-year-old man comes to the emergency department because of a 30-minute history of severe flank pain. It is accompanied by painful urination (dysuria) and blood in the urine (hematuria). The pain began yesterday as a vague ache but has since intensified. Since then, the pain has moved toward his lower abdomen and groin. It is currently a sharp, intermittent sensation with periods of worsening intensity that makes finding a comfortable position difficult. Imaging studies is shown. Which of the following is the most likely location of the obstruction?

A. Right ureteropelvic junction
B. Right pelvic brim
C. Right ureterovesical junction
D. Left ureteropelvic junction
E. Left pelvic brim
F. Left ureterovesical junction



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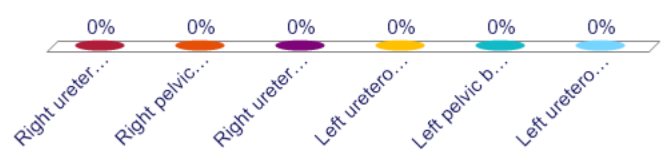

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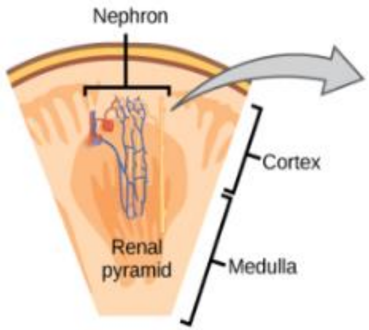
Case 2

A 22-year-old man comes to the emergency department because of a 30-minute history of severe flank pain. It is accompanied by painful urination (dysuria) and blood in the urine (hematuria). The pain began yesterday as a vague ache but has since intensified. Since then, the pain has moved toward his lower abdomen and groin. It is currently a sharp, intermittent sensation with periods of worsening intensity that makes finding a comfortable position difficult. Imaging studies is shown.

- Which side is affected?
- At what level is the ureteral obstruction located?

Pathway of Urine Flow



Flow Direction:

Renal papilla

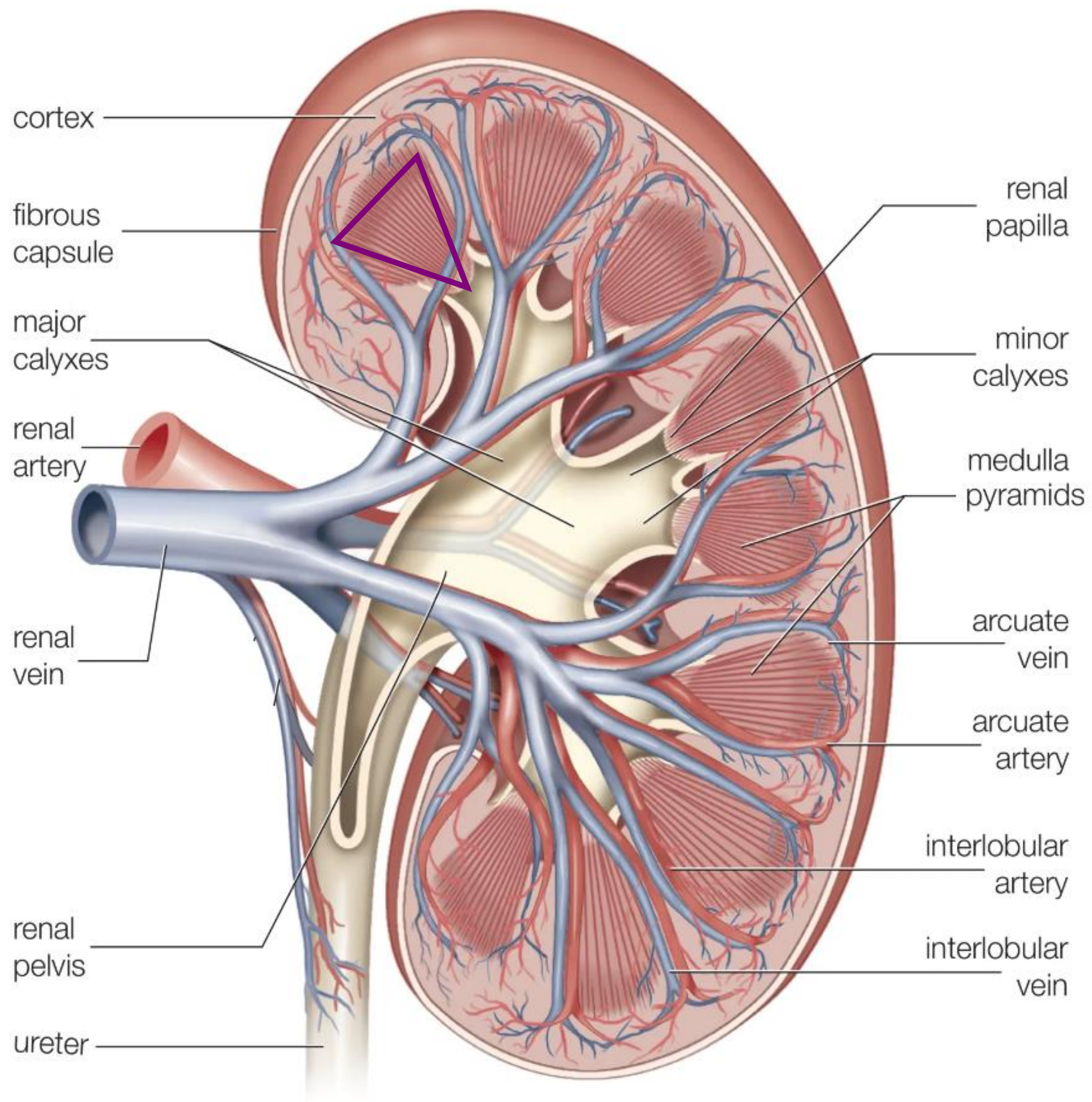
Minor calyces

Major calyces

Renal pelvis

Ureter

Urinary bladder



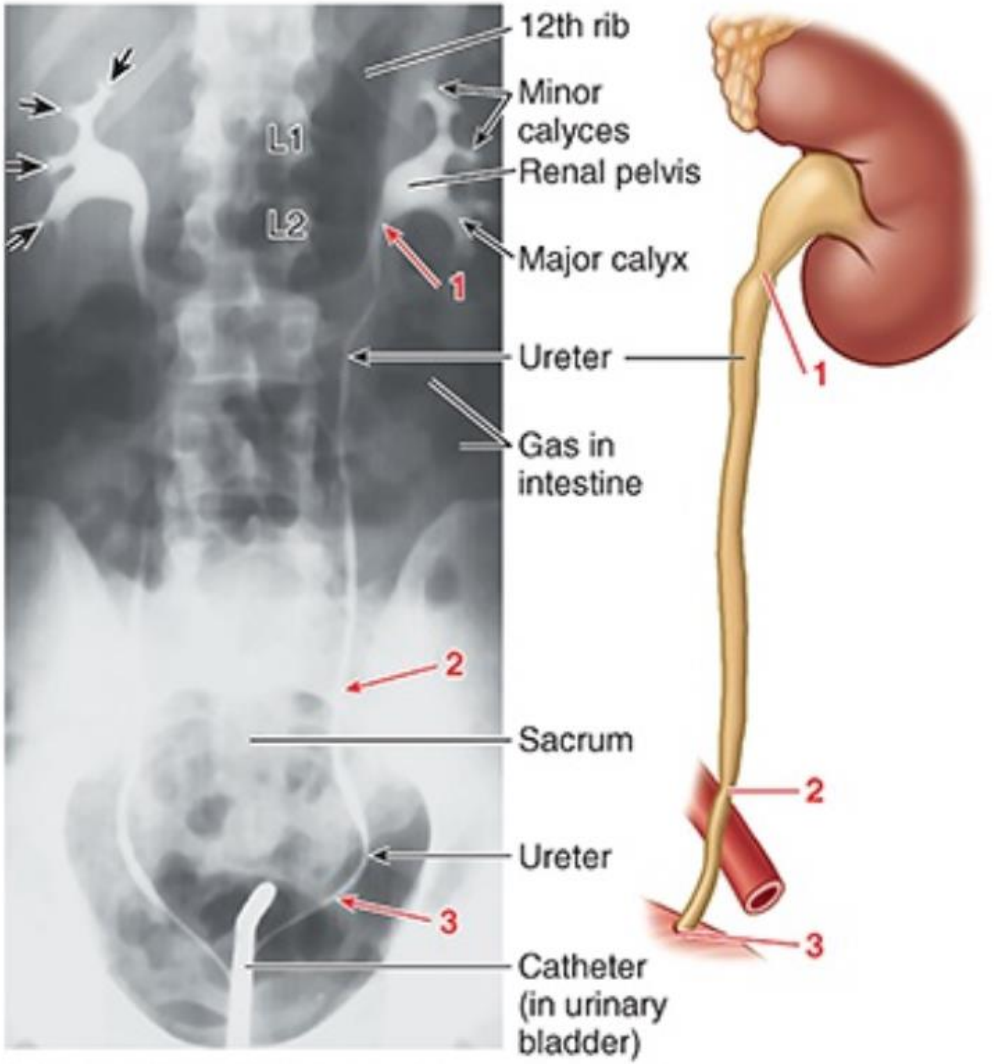
Ureteric Constrictions

Points of narrowest caliber in the ureter are:

1) Ureteropelvic junction: transition from renal pelvis to ureter

2) Pelvic brim: crossing the iliac vessels

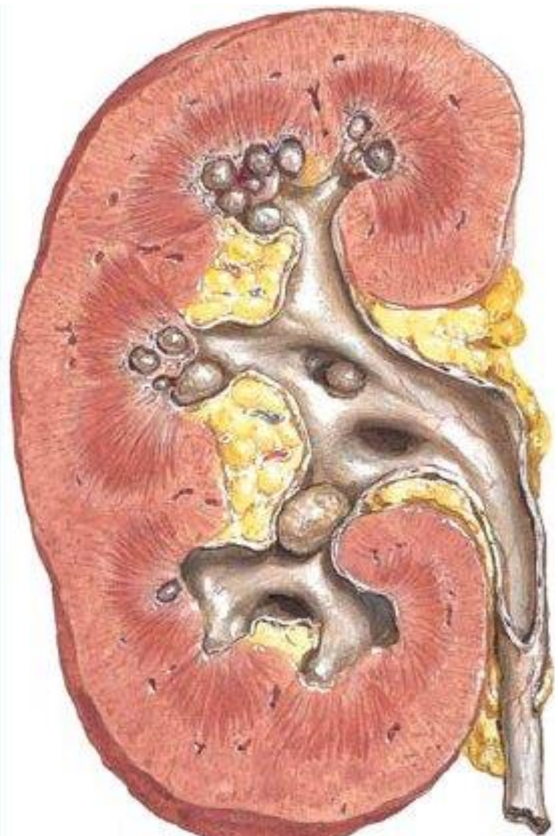
3) Ureterovesical junction: passing through the bladder wall



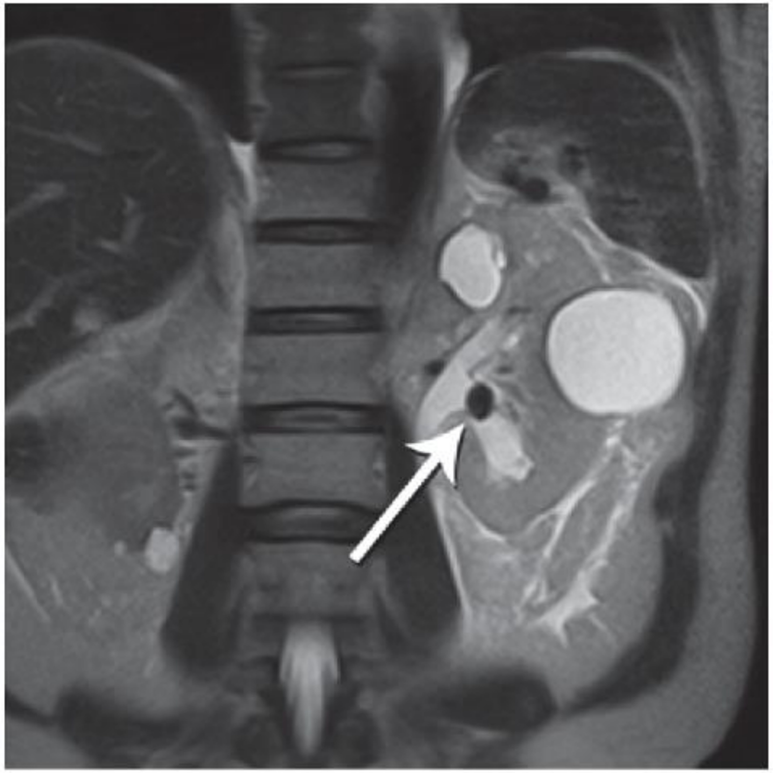
Ureteric Stones (Calculi)

Urolithiasis: solid deposits present in the urinary tract

- Often form initially in the renal pelvis (nephrolithiasis)
- Can move to within the ureter (ureterolithiasis)
- Can lead urinary tract obstruction at **narrow points**
  - Can lead to hydronephrosis

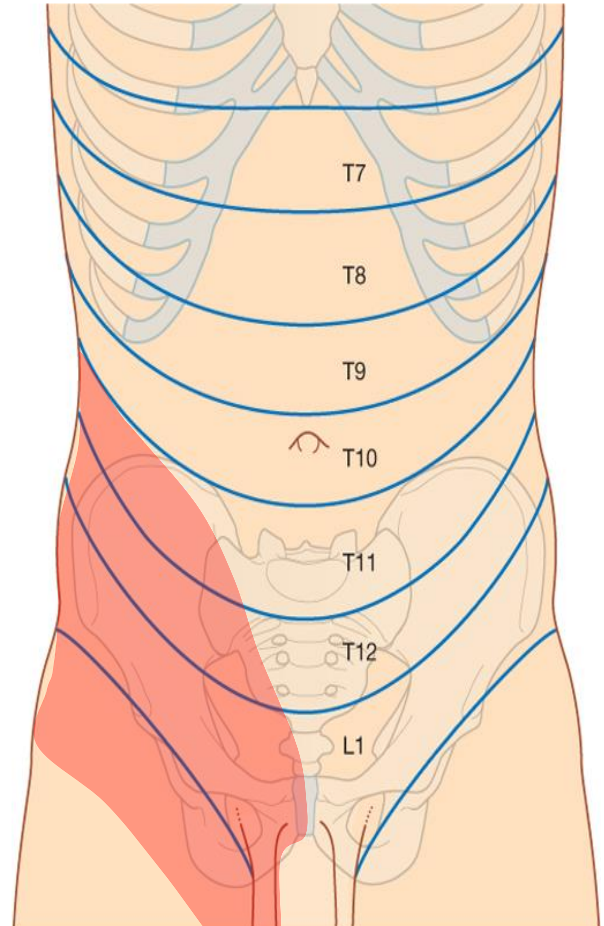


Multiple small calculi



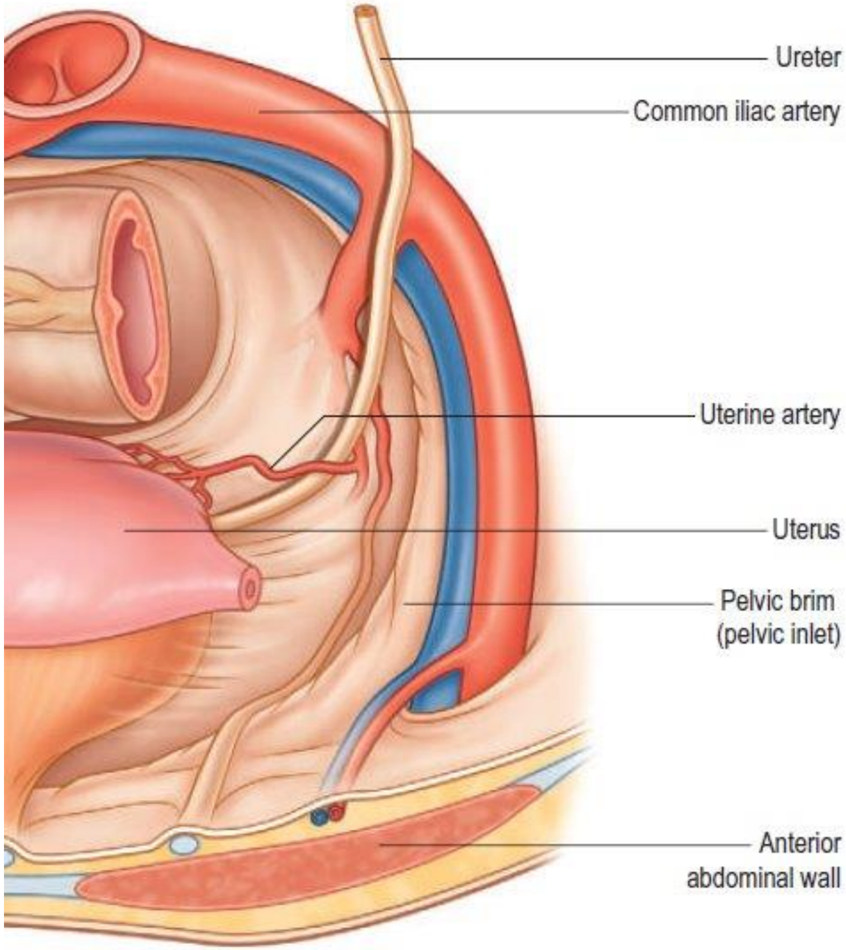
Ureteric (Renal) Colic

- Spasmodic, agonizing pain due to distension and contraction of ureteric muscle
  - i.e. obstructing stone is gradually forced down the ureter by muscle spasms
- Ureter is innervated from T11-L2 levels
  - "from loin to groin"
- Afferent fibers accompany sympathetics
  - pain is referred to cutaneous areas supplied by these spinal cord levels.
  - involvement with vagal fibers may lead to concomitant nausea and vomiting



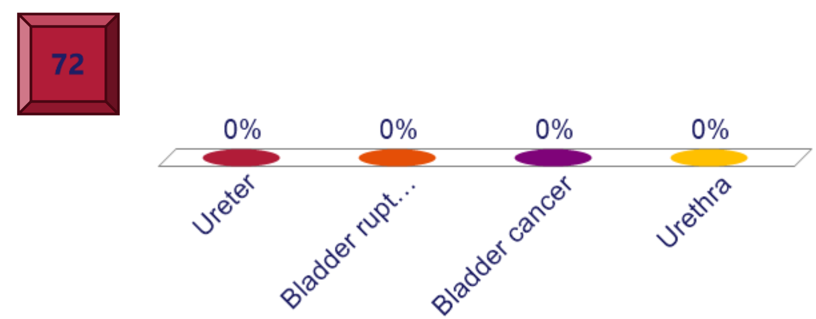
Hysterectomy & Urinary System

- During hysterectomy: possible ureteric injury
  - Most commonly occurs with the distal ureter at the level of the uterine artery
  - Located as "water under a bridge"
- After hysterectomy: possible worsening of urinary symptoms
  - e.g. urge incontinence
  - lower part of the bladder receives part of its blood supply from uterine artery



A 24-year-old man presents with a 2-week history of blood in his urine. A cystoscopy is performed under general anesthesia. Upon advancing the scope into the bladder, a triangular area on the bladder base is visualized, bordered by the internal opening of the urethra and two narrow, slit-shaped openings. Active bleeding is noted coming from one of these slit-like openings, which is surrounded by normal-appearing urothelium. No masses or mucosal lesions are seen in the bladder. Which of the following is the most likely source of the blood?

A. Ureter
B. Bladder rupture
C. Bladder cancer
D. Urethra



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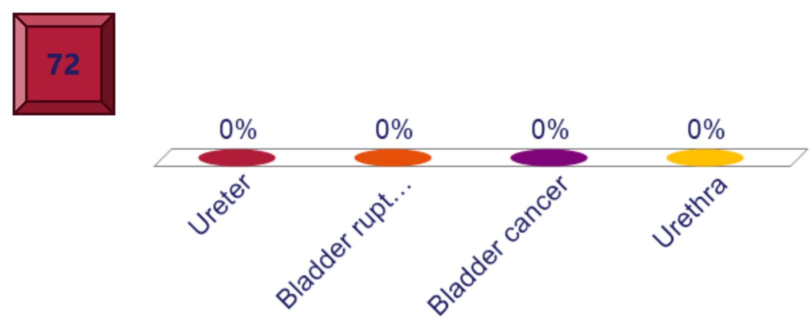

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- What are the possibly affected structures that would lead to this condition?
- What are the support structures in a normal bladder?

Urinary Bladder Anatomy

Body (DLA)

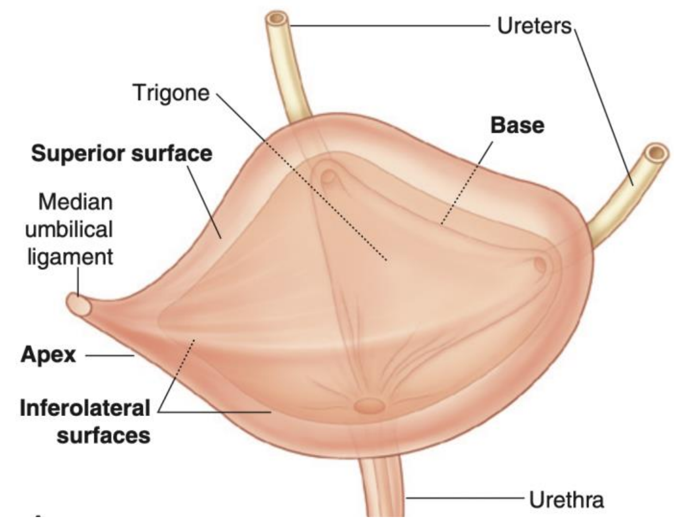
Apex (DLA)

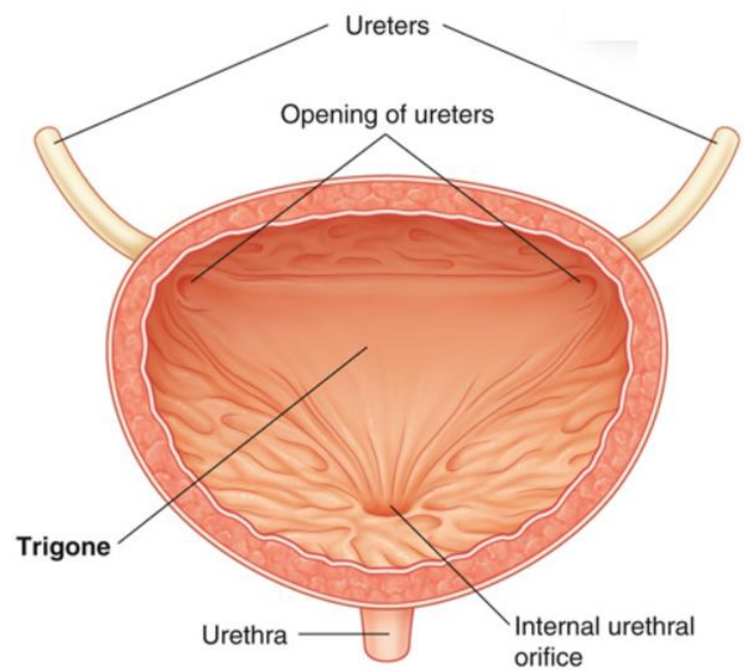
Fundus (Base)

- inverted triangle that faces posteroinferiorly
- two ureters enter at upper corners
- urethra drains from lower corner
- **Trigone**: smooth area between these openings
  - Bulk of trigone is derived from **detrusor** - bladder smooth muscle

Neck

- The lowest and most fixed part of the bladder.
- continuous with the urethra, which excretes urine outside of the body





Bladder Support

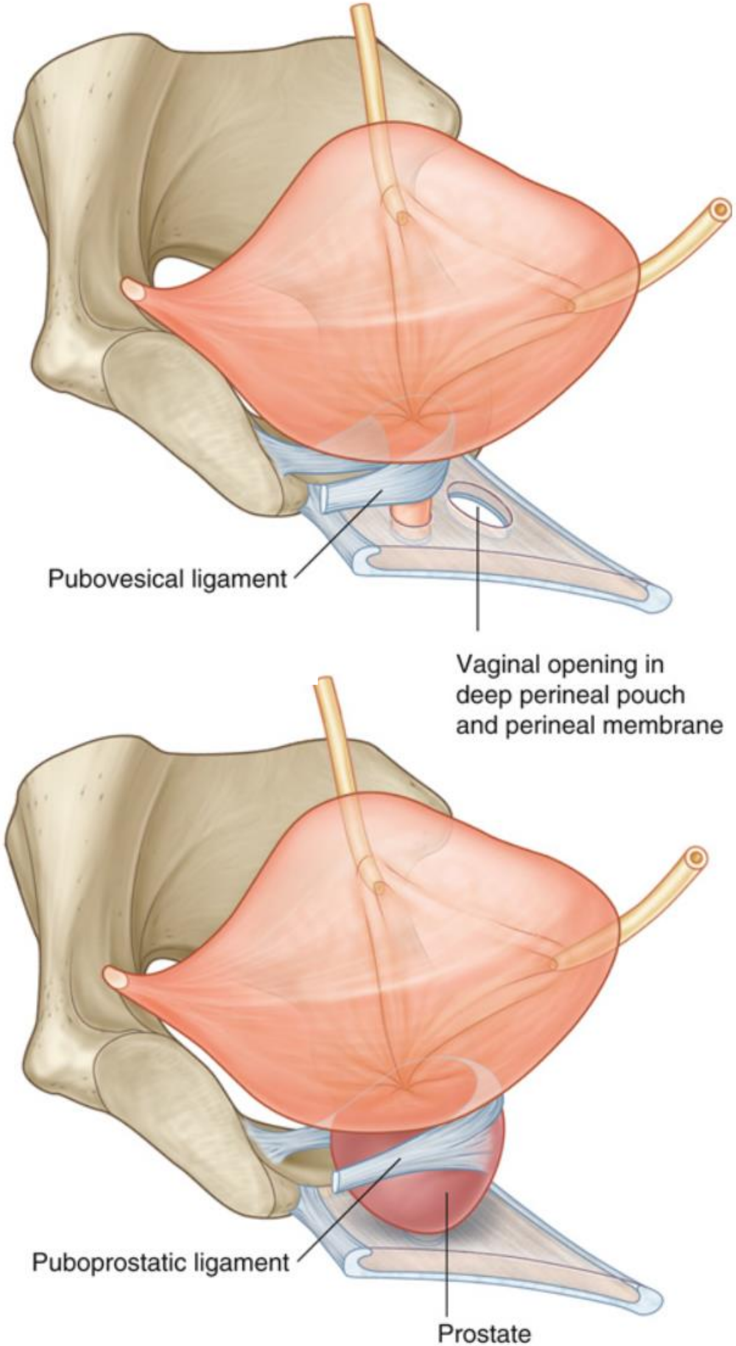
- The neck is stabilized by paired fibromuscular bands that connect it to the posteroinferior aspect of each pubic bone.

In women:

- The fibromuscular bands are called the Pubovesical ligament .
- They work with the **perineal membrane**, **levator ani muscles**, and **pubic bones** to support the bladder.

In men:

- The fibromuscular bands are called the Puboprostatic ligament
- They merge with the **fibrous capsule of the prostate**, which surrounds the bladder neck and adjacent urethra.



Cystocele

- Pubocervical fascia forms a critical layer between the bladder and the vagina; its integrity is essential in maintaining the normal position of the bladder.
- In cases of **childbirth trauma**, repeated childbirth **(multiparous women),** increased **intra-abdominal pressure**, or **atrophy** due to aging and estrogen loss, this fascia may weaken or tear.
- As a result, the bladder bulges into the vaginal canal, forming a cystocele.

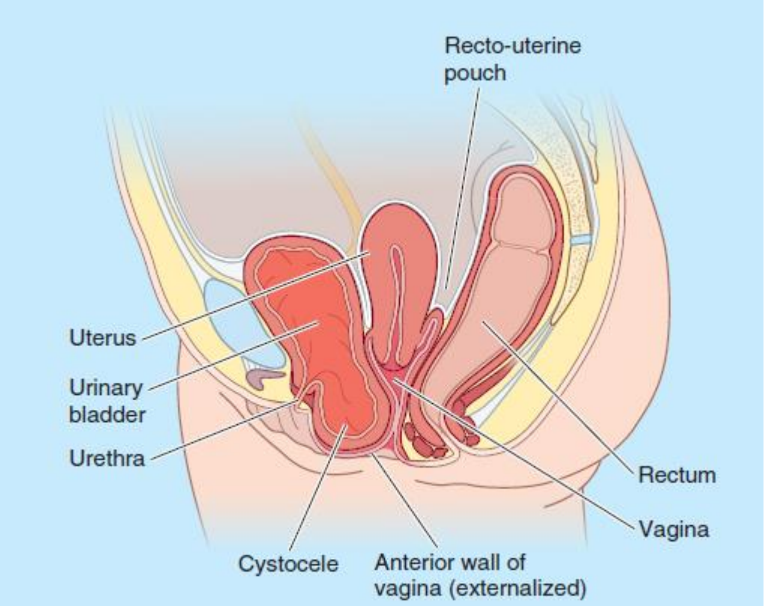


FIGURE B3.8.

A 76-year-old man is admitted to the hospital following surgery for a hip fracture. On postoperative day 2, he is unable to void despite a strong sensation of bladder fullness. A Foley catheter is inserted, and resistance is met just after passing the catheter through the membranous urethra. The catheter is then successfully advanced with gentle rotation and urine begins to drain. Through which of the following structures did the catheter pass immediately prior to entering the bladder and allowing urine to flow?

A. Membranous urethra
B. Prostatic urethra
C. Spongy (penile) urethra
D. External urethral sphincter
E. Bulb of the penis

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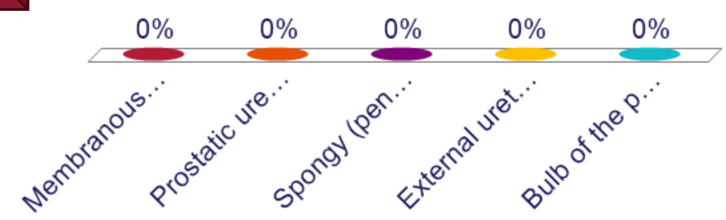

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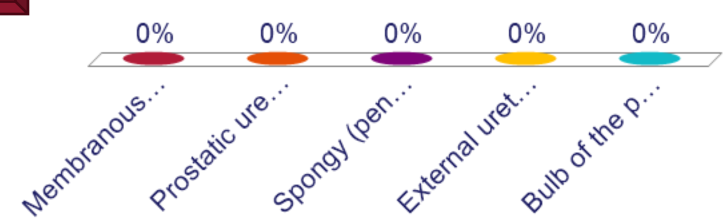

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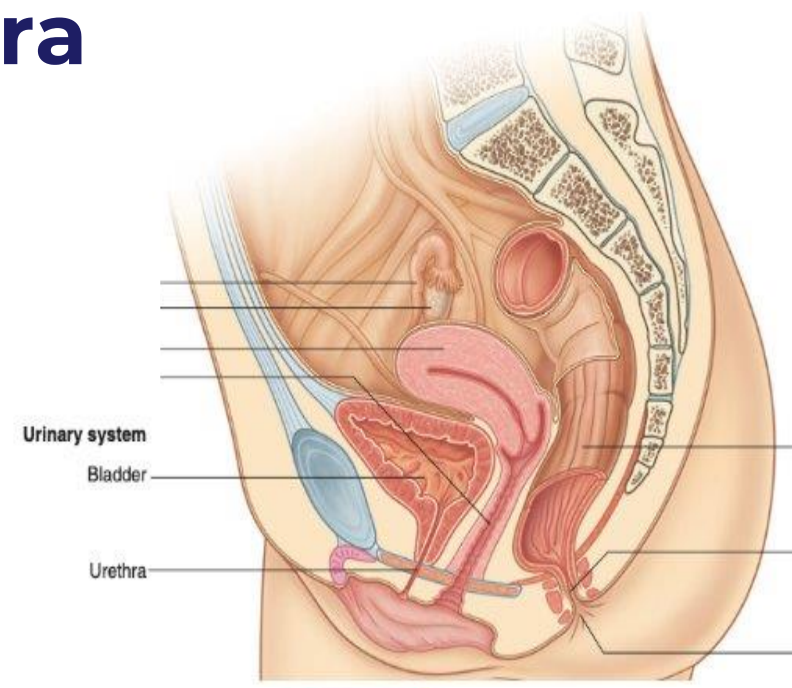
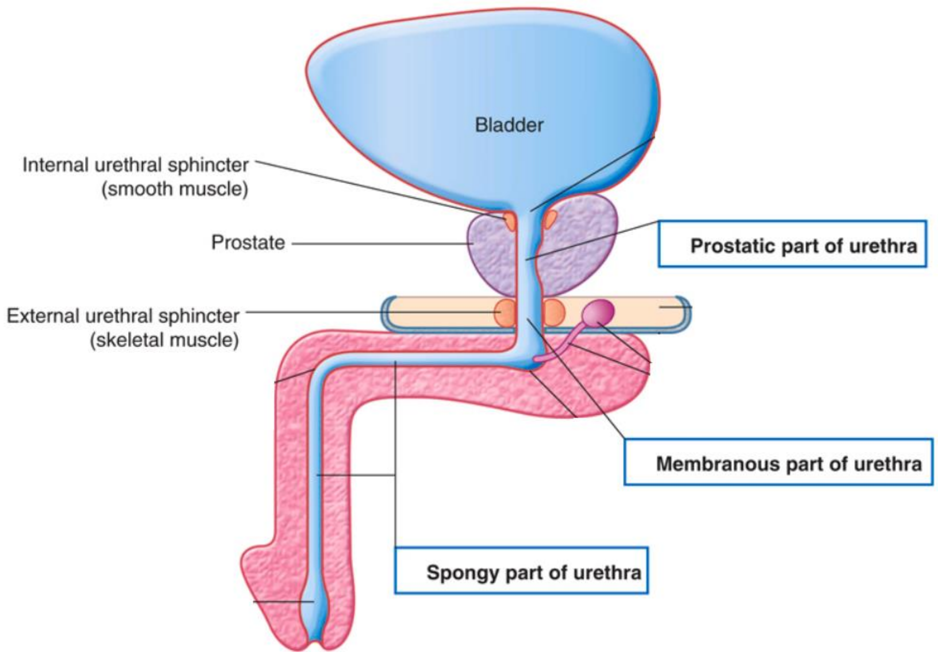
- Describe the anatomical structure of the male urethra and compare it to that of the female urethra.

Anatomy of the Urethra

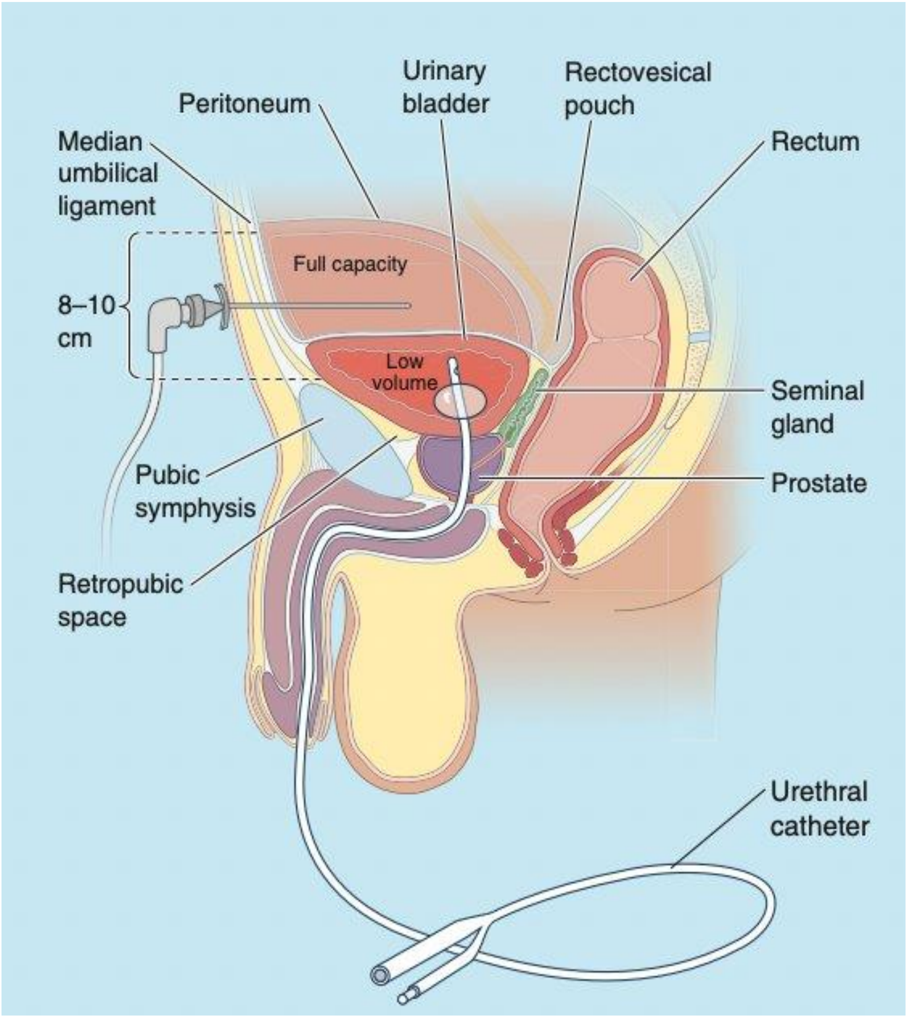
- Urethra: muscular tube that facilitates urine excretion
  - conveys urine from the internal urethral orifice (bladder) to the external urethral orifice (penis or vestibule)
  - Female urethra is shorter (4 cm) than male urethra (20 cm)
    - male urethra has 3 sections: prostatic, membranous, spongy
  - Internal urethral sphincter: Junction of bladder and urethra (Smooth muscle; Involuntary control)
  - External urethral sphincter: Membranous urethra in males; mid-urethra in females (deep perineal pouch) (Skeletal muscle; Voluntary Control)





Urinary Catheterization

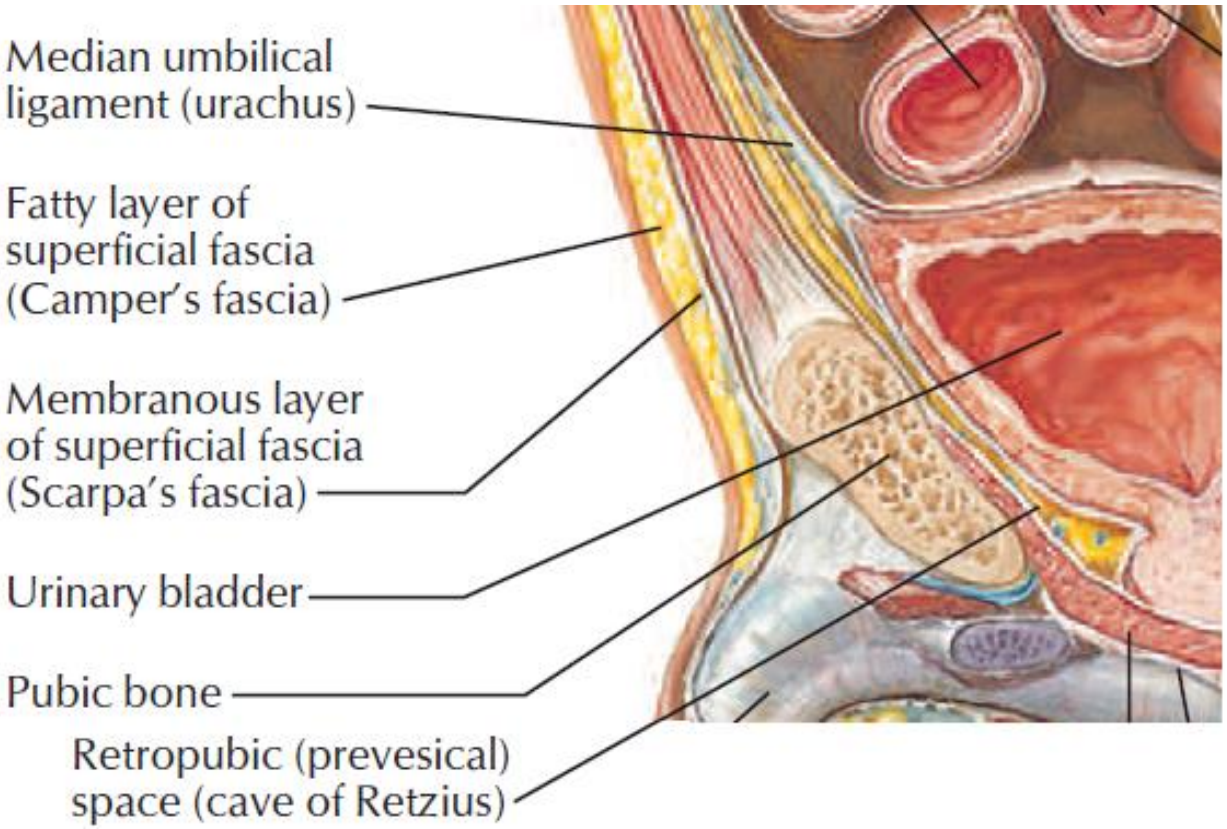
- **Internal urethral catheterization**
  - Artificial tube insertion through the urethra to the bladder
  - Done to remove urine or irrigate the bladder



- **Suprapubic catheterization**
  - The most invasive type of urinary catheter, inserted directly into the bladder through the abdominal wall.
  - When the bladder fills, it extends superiorly above the pubic symphysis
    - Bladder lies adjacent to abdominal wall without intervening peritoneum

Retropubic Space (of Retzius)

- Potential space between bladder and pubis
  - Pubic bones and pubic symphysis anteriorly
  - Bladder rests posteriorly on prostate (males) or anterior wall of the vagina (females)
- Surgically accessible by separating rectus abdominis at midline and dissecting towards peritoneum



A 44-year-old G5P5 woman comes to the physician because of a 6-month history of involuntary loss of urine. She has no history of pelvic surgery. Physical examination is unremarkable. Laboratory studies, including urinalysis and serum creatinine, are unremarkable. During evaluation, she is asked to stand with a comfortably full bladder and to cough forcefully. Leakage of urine is observed during this maneuver. Which of the following best explains the underlying mechanism of this patient's condition?

A. Increased tone of the external urethral sphincter
B. Overactivity of the detrusor muscle during bladder filling
C. Weakened support of the urethra by the pelvic floor muscles
D. Compression of the urethra by an enlarged uterus
E. Fibrosis of the bladder wall due to chronic inflammation

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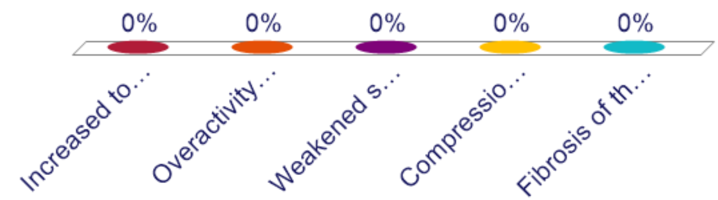

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A. Increased tone of the external urethral sphincter
B. Overactivity of the detrusor muscle during bladder filling
C. Weakened support of the urethra by the pelvic floor muscles
D. Compression of the urethra by an enlarged uterus
E. Fibrosis of the bladder wall due to chronic inflammation

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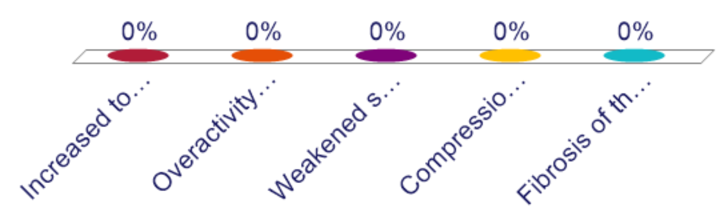

Case 5

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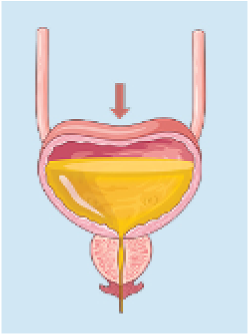
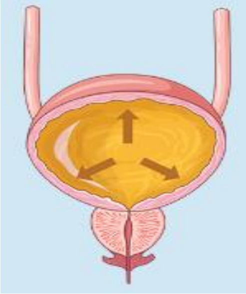
- What is the most likely diagnosis?
- What is the innervation of the bladder?

Types of Urinary Incontinence



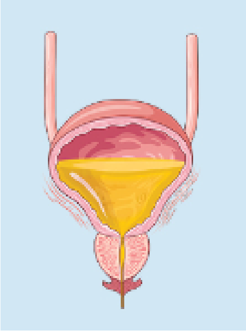
Stress incontinence

- Sudden rise in intra-abdominal pressure (e.g., coughing, sneezing) → Bladder pressure exceeds urethral closing pressure → urinary leakage
- Common cause: Urethral sphincter dysfunction
- Risk factor: Multiple prior vaginal deliveries → obstetric trauma → **urethral sphincter injury** or **pudendal nerve injury** Leads to weakened pelvic floor muscles and **decreased external urethral sphincter (EUS) tone**



Overflow incontinence

- Overfilled bladder that cannot fully empty
- due to weakened detrusor or outlet obstruction
- Incomplete emptying & persistent involuntary dribbling



Urge incontinence

- Sudden urges to micturate
- due to detrusor hyperactivity (involuntary contraction)

Bladder Innervation – Efferent

- Continence is maintained by
  1. Bladder (detrusor muscle)
  2. Internal urethral sphincter (IUS)
  3. External urethral sphincter (EUS)

1. External urethral sphincter (EUS): Skeletal muscle under voluntary control and is innervated by somatic motor (efferent) fibers via Pudendal nerve (S2-4).

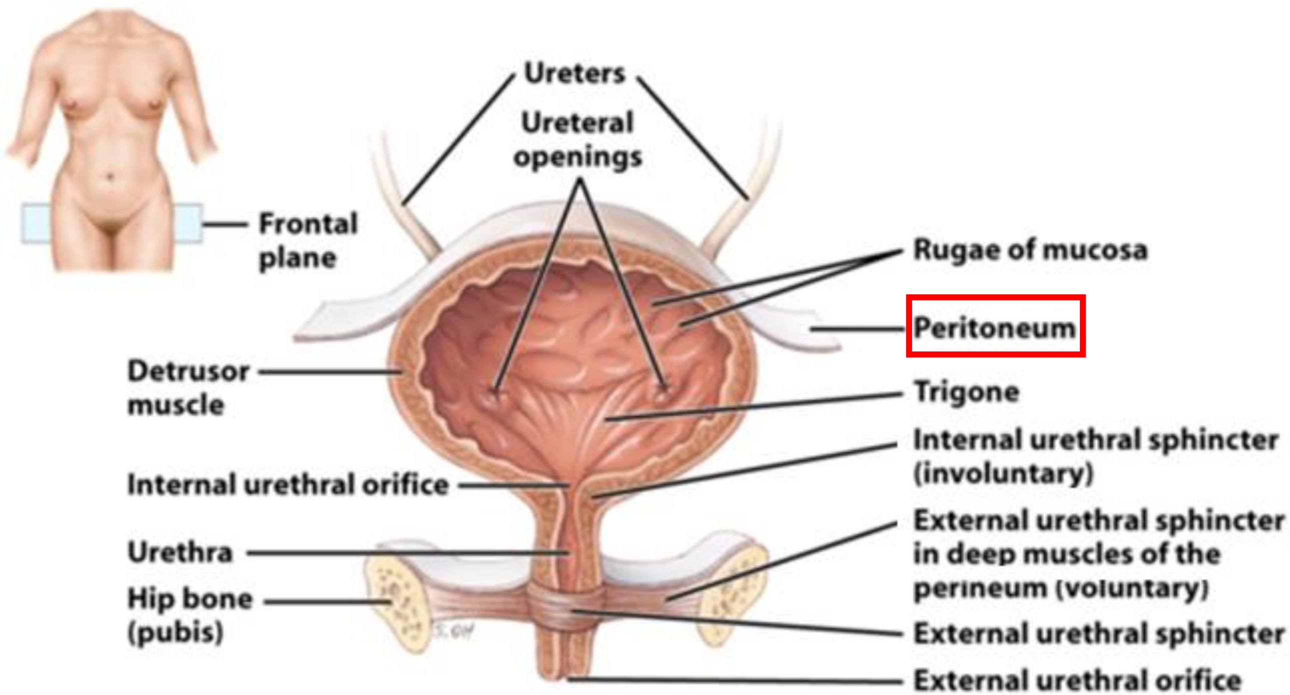
  - Cortical control is learned over time, e.g., infants automatically void

2. Bladder and **3. Internal urethral sphincter (IUS):** Smooth muscles under involuntary control by visceral motor (efferent) fibers (**autonomic** innervation)

- **Sympathetic** activity:
  - T10-12 (lesser, least splanchnics) and L1-2 (lumbar splanchnics)
  - **Increases IUS** tone
  - **Inhibits detrusor** contraction → promotes urine **storage**
- **Parasympathetic** activity:
  - from S2-4 (pelvic splanchnics)
  - **Relaxes IUS** tone
  - **Stimulates detrusor** contraction → promotes **micturition**

Bladder Innervation – Afferent

- Visceral afferents for pain travel with *sympathetic/parasympathetic* nerves
    - **Pelvic pain line (PPL)** is an imaginary line that corresponds to the lower limit of the peritoneum. It represents a boundary between sympathetic and parasympathetic paths
    - **Above PPL (peritoneal portion of bladder)** - the visceral afferents for pain travel via the sympathetics
    - **Below PPL (majority of the urinary bladder)** - the visceral afferents for pain travel via the parasympathetics.
- Visceral afferents for sensation of fullness/stretch travel primarily with *parasympathetic* nerves

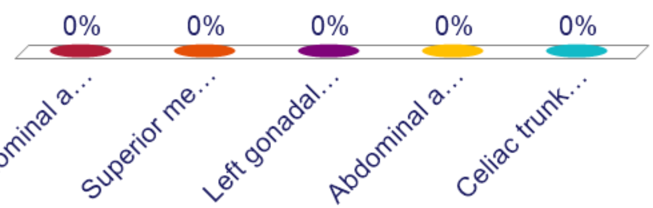


Anterior view of frontal section

A 22-year-old man comes to the clinic with a 3-month history of intermittent left-sided flank pain and visible blood in his urine. He reports that the pain worsens after prolonged standing or physical activity. He has no history of trauma, stones, or urinary tract infections. Urinalysis confirms the presence of hematuria. A Doppler ultrasound reveals compression of the left renal vein as it passes between two major vessels. Compression of the left renal vein in this condition is most likely occurring between which of the following two structures?

A. Abdominal aorta and inferior vena cava
B. Superior mesenteric artery and abdominal aorta
C. Left gonadal vein and ureter
D. Abdominal aorta and left psoas major muscle
E. Celiac trunk and splenic vein

72



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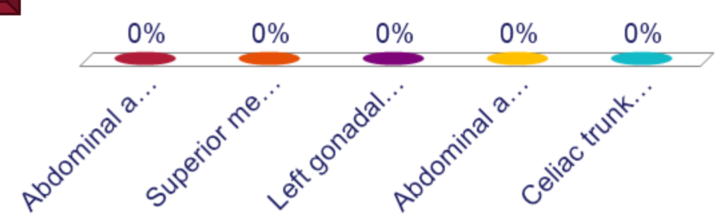

Case 6

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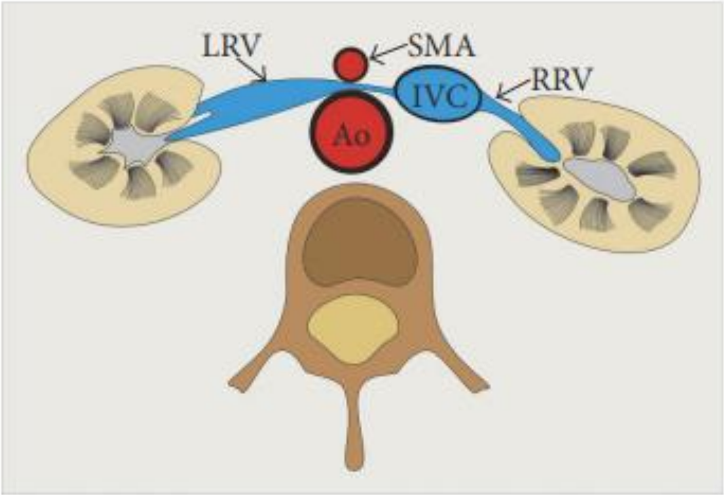
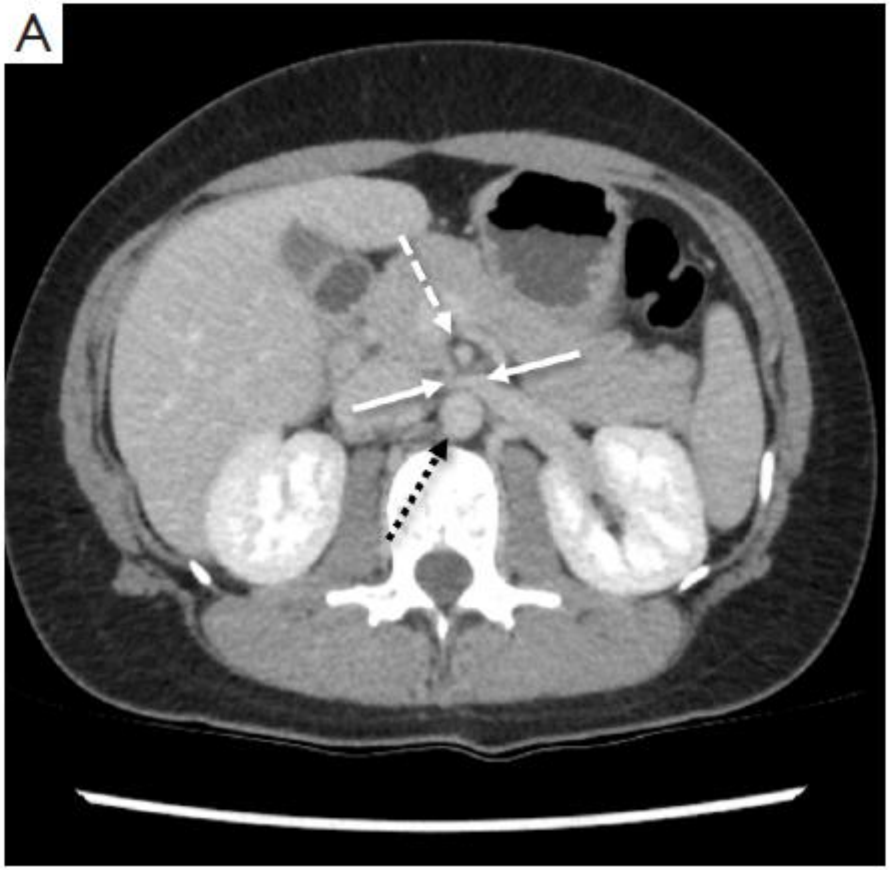
- What is renal vein entrapment syndrome "nutcracker syndrome"?
- Which blood vessels are most likely involved?

Renal Vein Entrapment Syndrome

- Nutcracker syndrome is a vascular compression disorder that refers to the entrapment and compression of the left renal vein.
    - Most commonly occurs between the superior mesenteric artery (SMA) and aorta



doi: 10.1155/2017/1746570

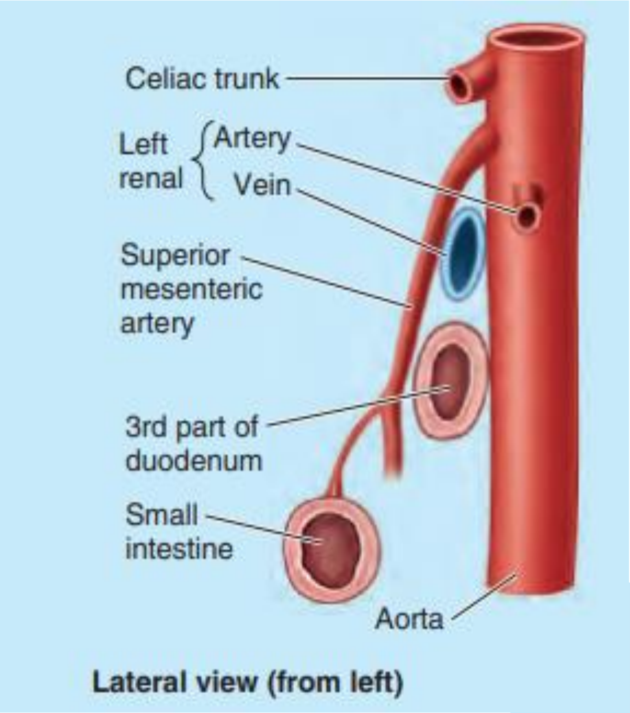


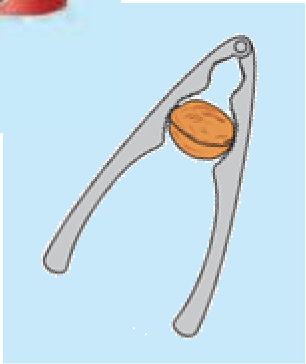
doi: 10.21037/cdt-20-160

Effects of Mesoaortic Compression

- Clinical manifestations include left flank pain, pelvic pain, gonadal varices, and hematuria (microscopic or macroscopic).

doi: 10.21037/cdt-20-160





A 42-year-old man comes to the physician because of a 2-month history of worsening dull flank pain and intermittent hematuria. He reports that his father died of renal failure at age 50. His blood pressure is 158/96 mm Hg. Abdominal examination reveals bilateral, ballotable flank masses. Ultrasound is shown. Which of the following is the most likely anatomical diagnosis in this patient?

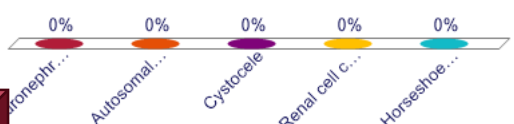
A. Hydronephrosis due to ureteropelvic junction obstruction

B. Autosomal dominant polycystic kidney disease

C. Cystocele

D. Renal cell carcinoma

E. Horseshoe kidney



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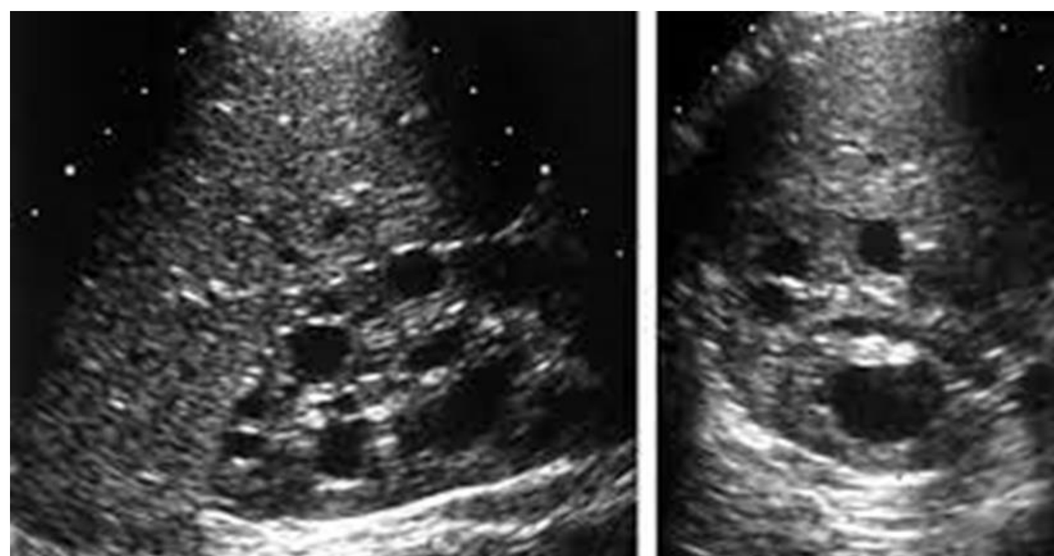

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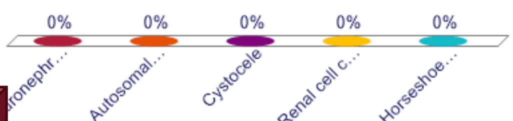

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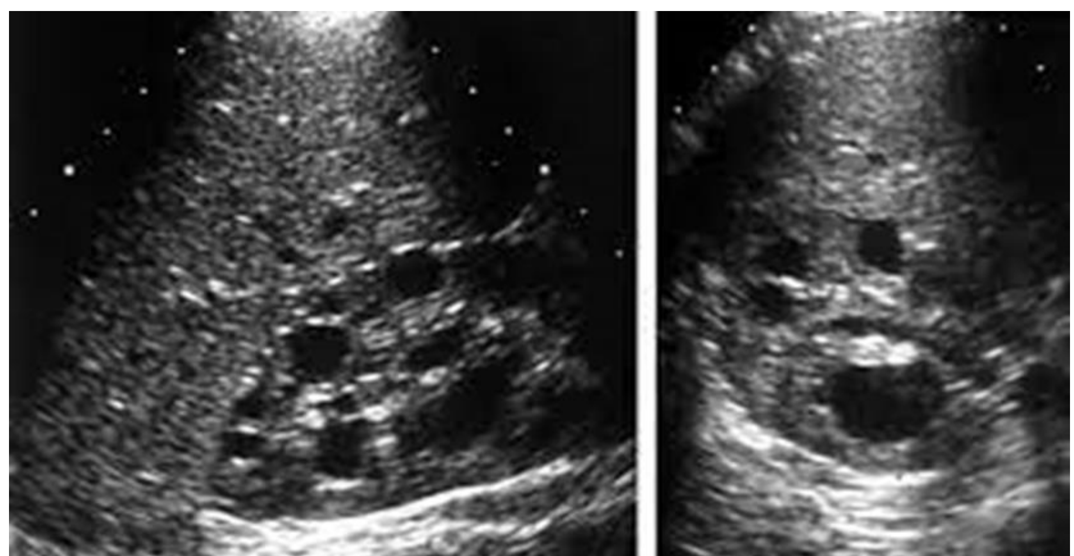

Case 7

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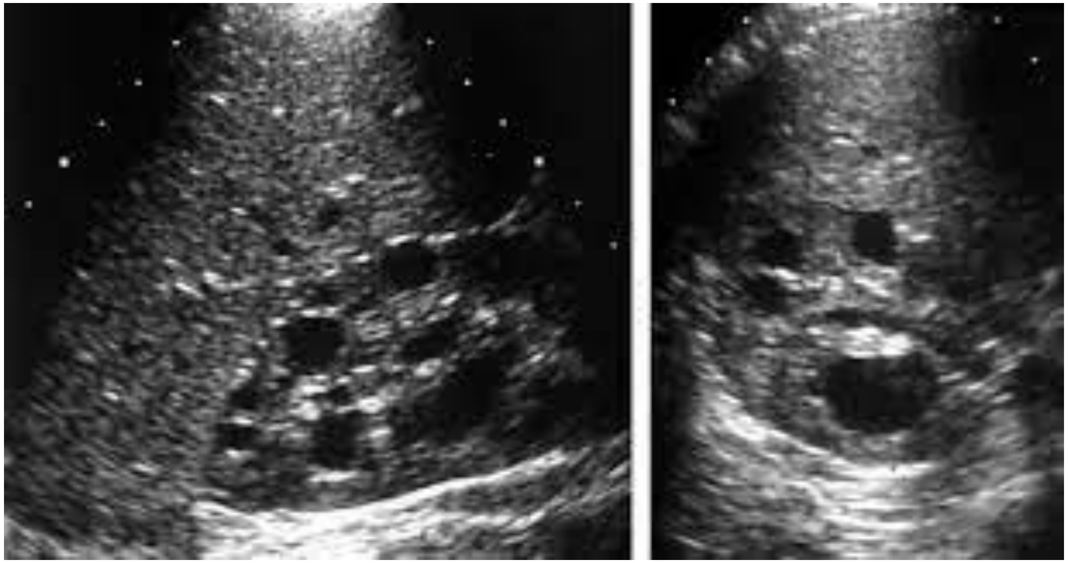
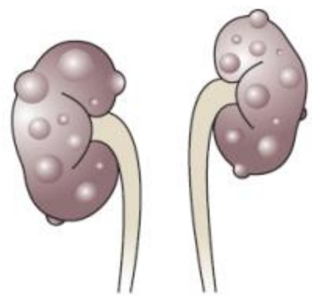
- What is the most likely diagnosis?
- Name conditions associated with this diagnosis.



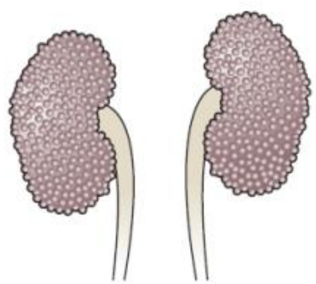
Polycystic Kidney

Inherited conditions with enlarging renal cysts



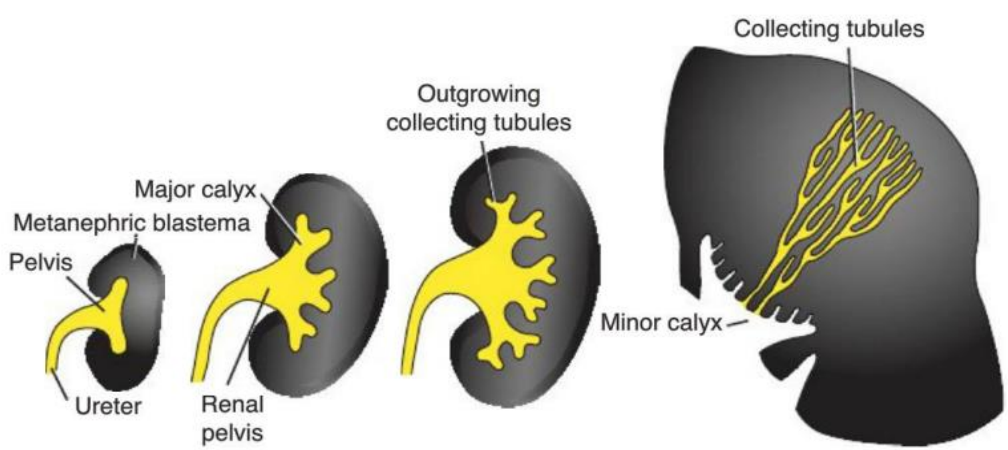
Autosomal dominant (more common)

- Cysts form from all segments of the nephron
- Approximately 50% of individuals will have end-stage kidney disease by 60 yo



Autosomal recessive

- Cysts progressively form as small dilations that expand in collecting tubules
- Kidneys become large and renal failure occurs in infancy or childhood



DOI: 10.1002/bdrb.10013

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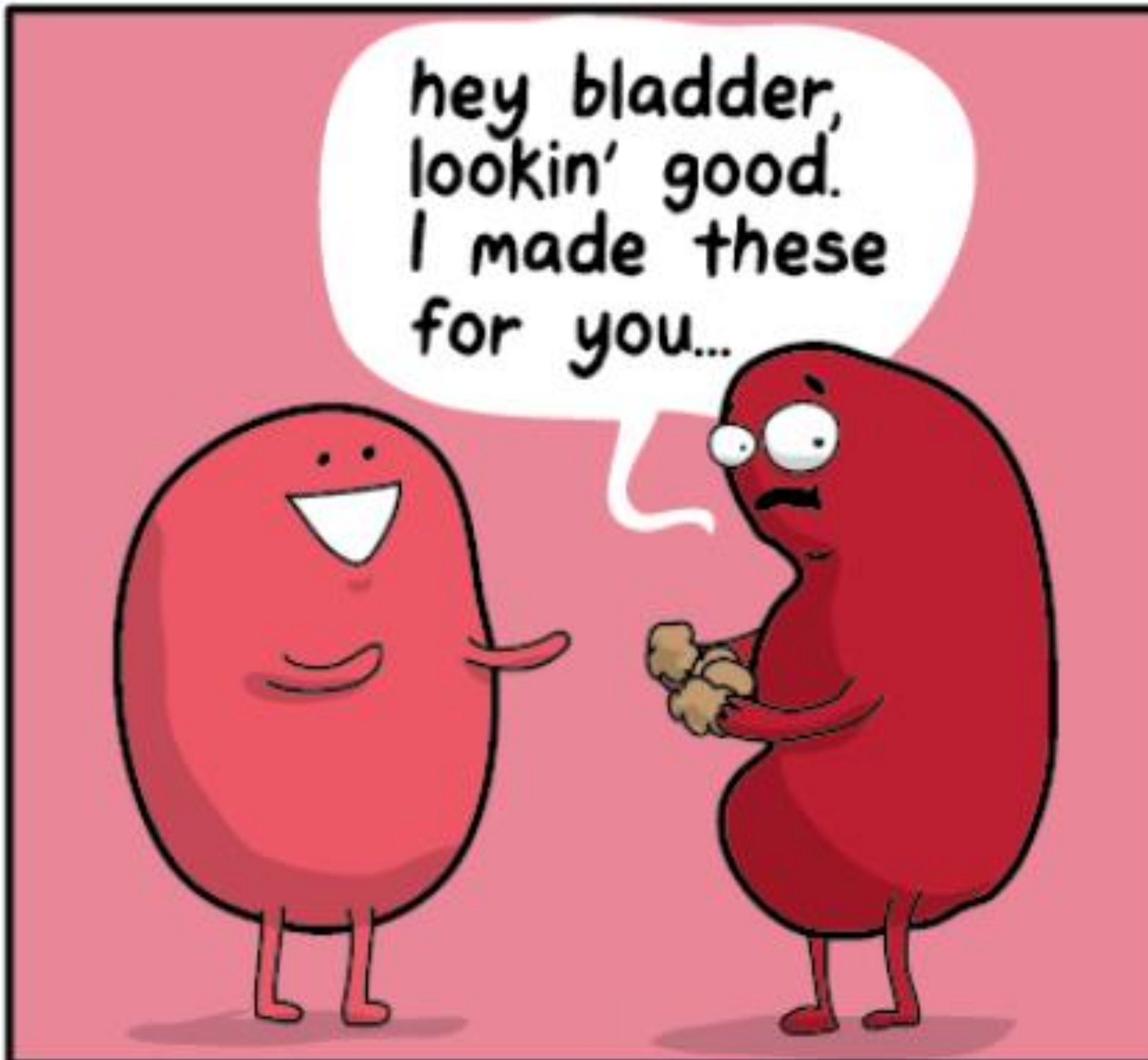
hey bladder, lookin' good. I made these for you...
theAwkwardYeti.com